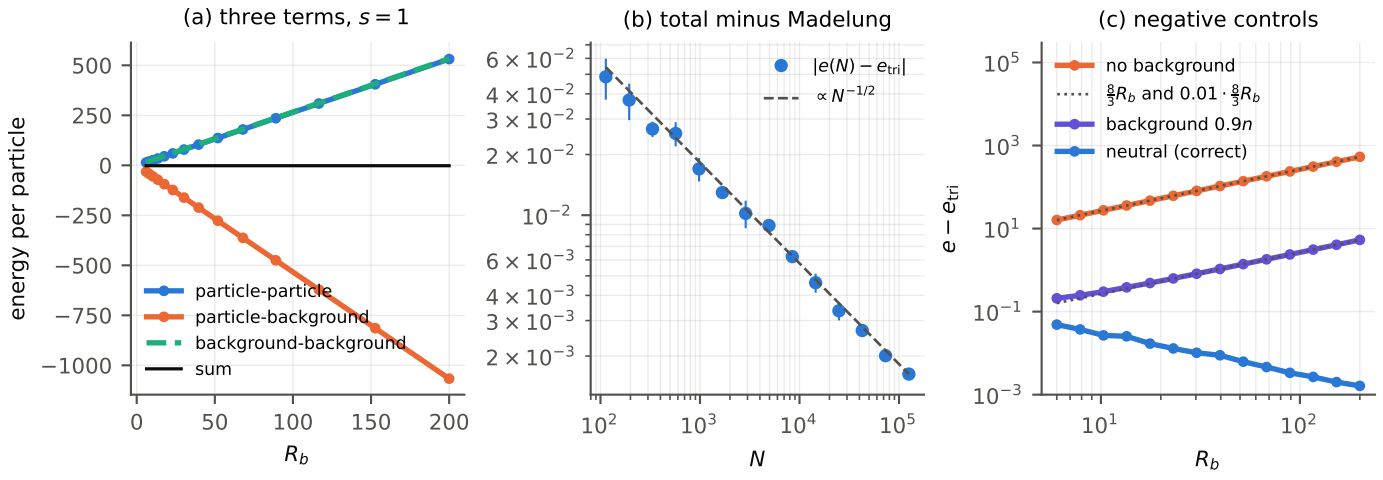
(a) three terms, $s = 1$
energy per particle
500
250
0
−250
−500
−750
−1000
0
50
100
150
200
R_b
particle-particle
particle-background
background-background
sum
(b) total minus Madelung
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^2
10^3
10^4
10^5
N
$|e(N) - e_{\text{tri}}|$
$\propto N^{-1/2}$
(c) negative controls
$e - e_{\text{tri}}$
10^5
10^3
10^1
10^{-1}
10^{-3}
10^1
10^2
R_b
no background
$\frac{8}{3}R_b$ and $0.01 \cdot \frac{8}{3}R_b$
background $0.9n$
neutral (correct)